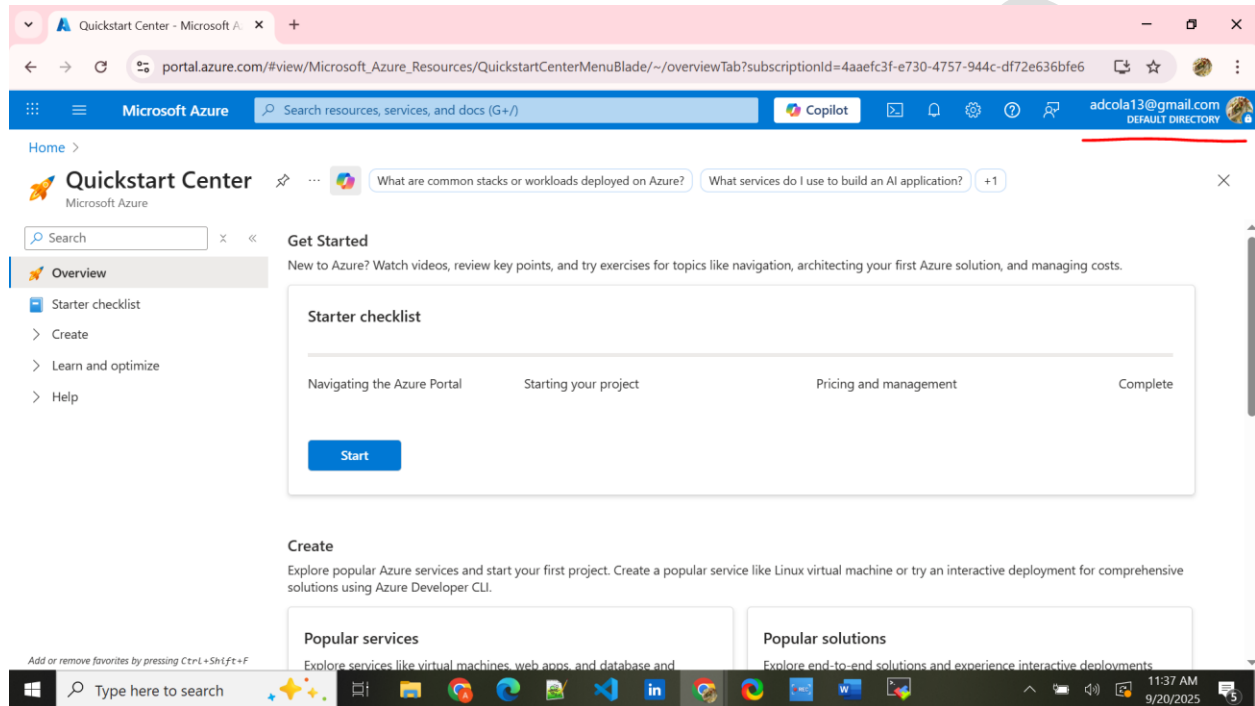


AZURE CLOUD

Assignment 23 – Create and Set Up Your Azure Free Account

Azure Portal Home



Subscription Overview

Microsoft Azure portal showing the overview of an Azure subscription (Subscription ID: 4aaefc3f-e730-4757-944c-df72e63bfe6).

Essentials

- Subscription ID: 4aaefc3f-e730-4757-944c-df72e63bfe6
- Directory: Default Directory (adcola13gmail.onmicrosoft.com)
- Status: Active
- Parent management group: 6e357e34-1d9d-4755-aea0-13cd3ef3ddda
- Subscription name: Azure subscription 1
- My role: Owner
- Plan: Azure Plan
- Secure Score: Not available

Spending rate and forecast

No data to display

Current cost: 0.00, Forecast: 0.00

Costs by resource

No active resource emitted usage yet.

Top free services by usage

Service	Usage
Azure Cosmos DB, Free Data Stored	0 / 25 (1 GB/Month)
Azure Cosmos DB, Free 100 RU/s	0 / 2976 (1/Hour)
Storage, Files, LRS Data Stored	0 / 100 (1 GB/Month)
Storage, Files, LRS Data Stored	0 / 100 (1 GB/Month)
Storage, Premium Page Blob, P6 Disks	0 / 2.2 (1/Month)
Storage, Standard HDD Managed Disks, LRS Snapshots	0 / 1 (1 GB/Month)
Storage, Standard HDD Managed Disks, Disk Operations	0 / 200 (10K)

Top products by number of resources

You don't have any resources in this subscription

Azure Defender coverage

Azure Defender is not enabled for this subscription

[Upgrade coverage](#)

[View all free services](#)

Assignment 24 – Deploy a React Application on an Azure Virtual Machine

1. Create a Resource Group

Microsoft Azure portal showing the details of a newly created Resource Group named **ReactAppRG1**.

Resources

Name	Type	Location
ReactAppUbuntu1	Virtual machine	Australia East
ReactAppUbuntu1-ip	Public IP address	Australia East
ReactAppUbuntu1-nsg	Network security group	Australia East
ReactAppUbuntu1-vnet	Virtual network	Australia East
reactappubuntu1135_x2	Network interface	Australia East
ReactAppUbuntu1_key	SSH key	Australia East
ReactAppUbuntu1_OsDisk_1_58a3cba896c54c9a206fea7c3a0cdc	Disk	Australia East

2. Provision an Ubuntu on Azure VM

The image shows two screenshots from the Azure portal. The top screenshot displays the Azure Marketplace with various operating system options. A red arrow points to the 'Ubuntu Server 22.04 LTS' option. The bottom screenshot shows the 'ReactAppUbuntu1' VM overview page. A red arrow points to the 'Networking' section, specifically the 'Public IP address' field, which shows '20.5.136.79'.

Top Screenshot: Azure Marketplace

Search the Marketplace:

Pricing: All | Operating System: All | Publisher Type: All | Product Type: All | Publisher name: All

☐ Azure benefit eligible only ☐ Azure services only

Popular Marketplace products

- Windows Server
- Windows 11
- Ubuntu Server 22.04 LTS
- SQL Server 2022 on Windows Server 2022
- Red Hat Enterprise Linux (RHEL) for Microsoft Azure
- Visual Studio 2022 (Microsoft Dev Box)

Trending now

Bottom Screenshot: ReactAppUbuntu1 VM Overview

Home > CreateVm-canonical.0001-com-ubuntu-server-jammy-2-20251001200451 | Overview >

ReactAppUbuntu1

Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search:

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Open in mobile | Feedback | CLI / PS

Essentials

Resource group (move): [ReactAppRG1](#)

Status: Running

Location: Australia East (Zone 2)

Subscription (move): [Azure subscription 1](#)

Subscription ID: 4aaefc3f-e730-4757-944c-df72e636bfe6

Availability zone: 2

Tags (edit): [Add tags](#)

Operating system: Linux (ubuntu 22.04)

Size: Standard B1s (1 vcpu, 1 GiB memory)

Primary NIC public IP: [20.5.136.79](#) [1 associated public IPs](#)

Virtual network/subnet: [ReactAppUbuntu1-vnet/default](#)

DNS name: [Not configured](#)

Health state: -

Time created: 1/10/2025, 2:41 pm UTC

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Virtual machine

Computer name: ReactAppUbuntu1

Operating system: Linux (ubuntu 22.04)

VM generation: V2

VM architecture: x64

Networking

Public IP address: [20.5.136.79](#) (Network interface reactappubuntu1135_z2) [1 associated public IPs](#)

Public IP address (IPv6): -

Private IP address: 10.0.0.4

3. Connect to the VM

```
2025-10-01 20:04:47 /drives/c/MobaXterm Portable v25.0 ssh -i ReactAppUbuntu1_key.pem azureuser@20.5.136.79
Warning: Permanently added '20.5.136.79' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1031-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Oct 1 14:43:57 UTC 2025

System load: 0.56          Processes:            109
Usage of /:  5.4% of 28.89GB Users logged in:       0
Memory usage: 30%         IPv4 address for eth0: 10.0.0.4
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

/usr/bin/xauth: file /home/azureuser/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

1. Install Node.js & npm

1. sudo apt update
2. sudo apt install -y nodejs npm
3. node -v && npm -v

```
azureuser@ReactApp:~$ node -v
v12.22.9
azureuser@ReactApp:~$ npm -v
8.5.1
azureuser@ReactApp:~$
```

2. Install and Start Nginx

1. sudo apt install -y nginx
2. sudo systemctl start nginx
3. sudo systemctl enable nginx

```
azureuser@ReactAppUbuntu1:~$ sudo systemctl start nginx
azureuser@ReactAppUbuntu1:~$ sudo systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nginx
```

3. Clone the React App Repo

1. git clone <https://github.com/pravinmishraaws/my-react-app.git>
2. cd my-react-app

```
azureuser@ReactAppUbuntu1:~$ git clone https://github.com/pravinmishraaws/my-react-app.git
Cloning into 'my-react-app'...
remote: Enumerating objects: 69, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (66/66), done.
remote: Total 69 (delta 28), reused 26 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (69/69), 188.95 KiB | 3.37 MiB/s, done.
Resolving deltas: 100% (28/28), done.
azureuser@ReactAppUbuntu1:~$ cd my-react-app/
```

4. Open the App.js file

- a. Navigate to your React app's source folder:

1. cd my-react-app/src

- b. Open the App.js file in a text editor:

1. nano App.js

(or use vi/vim if you prefer)

- c. **Modify the content**

<h2>Deployed by: Your Full Name</h2>

<p>Date: DD/MM/YYYY</p>

Update your details like: Your Full Name & Date

Build Application for Production:

1. Install Dependencies & Build App

1. npm install
2. npm run build

```
azureuser@ReactAppUbuntu1:~/my-react-app$ npm run build

> my-react-app@0.1.0 build
> react-scripts build

Creating an optimized production build...
Browserslist: browsers data (caniuse-lite) is 8 months old. Please run:
  npx update-browserslist-db@latest
  Why you should do it regularly: https://github.com/browserslist/update-db#readme
Compiled successfully.

File sizes after gzip:

 59.21 kB  build/static/js/main.e940c00a.js
 1.78 kB   build/static/js/453.ad6eb26e.chunk.js
264 B     build/static/css/main.e6c13ad2.css

The project was built assuming it is hosted at /.
You can control this with the homepage field in your package.json.

The build folder is ready to be deployed.
You may serve it with a static server:

  npm install -g serve
  serve -s build

Find out more about deployment here:

https://cra.link/deployment
```

2. Serve React App via Nginx

1. `sudo rm -rf /var/www/html/*`
2. `sudo cp -r build/* /var/www/html/`
3. `sudo chown -R www-data:www-data /var/www/html`
4. `sudo chmod -R 755 /var/www/html`

3. Configure Nginx

Replace the default config:

```
echo 'server {
    listen 80;
    server_name _;
    root /var/www/html;
    index index.html;

    location / {
```

```
try_files $uri /index.html;  
}
```

```
error_page 404 /index.html;  
}' | sudo tee /etc/nginx/sites-available/default > /dev/null
```

sudo systemctl restart nginx

```
azureuser@ReactAppUbuntu1:~/my-react-app$ sudo rm -rf /var/www/html/*  
azureuser@ReactAppUbuntu1:~/my-react-app$ sudo cp -r build/* /var/www/html/  
azureuser@ReactAppUbuntu1:~/my-react-app$ sudo chown -R www-data:www-data /var/www/html  
azureuser@ReactAppUbuntu1:~/my-react-app$ sudo chmod -R 755 /var/www/html  
azureuser@ReactAppUbuntu1:~/my-react-app$ echo 'server {  
    listen 80;  
    server_name _;  
    root /var/www/html;  
    index index.html;  
  
    location / {  
        try_files $uri /index.html;  
    }  
  
    error_page 404 /index.html;  
}' | sudo tee /etc/nginx/sites-available/default > /dev/null  
azureuser@ReactAppUbuntu1:~/my-react-app$ sudo systemctl restart nginx  
azureuser@ReactAppUbuntu1:~/my-react-app$ curl ifconfig.me  
20.5.136.79  
azureuser@ReactAppUbuntu1:~/my-react-app$ history
```

4. Test the App

Get your public IP:

1. curl ifconfig.me

```
azureuser@ReactAppUbuntu1:~/my-react-app$ curl ifconfig.me
20.5.136.79
azureuser@ReactAppUbuntu1:~/my-react-app$ history
 1  sudo apt update
 2  sudo apt upgrade
 3  sudo apt install -y nodejs npm
 4  node -v
 5  npm -v
 6  sudo apt install -y nginx
 7  sudo systemctl start nginx
 8  sudo systemctl enable nginx
 9  git clone https://github.com/pravinmishraaws/my-react-app.git
10  cd my-react-app/
11  cd src/
12  nano App.js
13  cd ..
14  npm install
15  npm run build
16  sudo rm -rf /var/www/html/*
17  sudo cp -r build/* /var/www/html/
18  sudo chown -R www-data:www-data /var/www/html
19  sudo chmod -R 755 /var/www/html
20  echo 'server {
listen 80;
server_name _;
root /var/www/html;
index index.html;

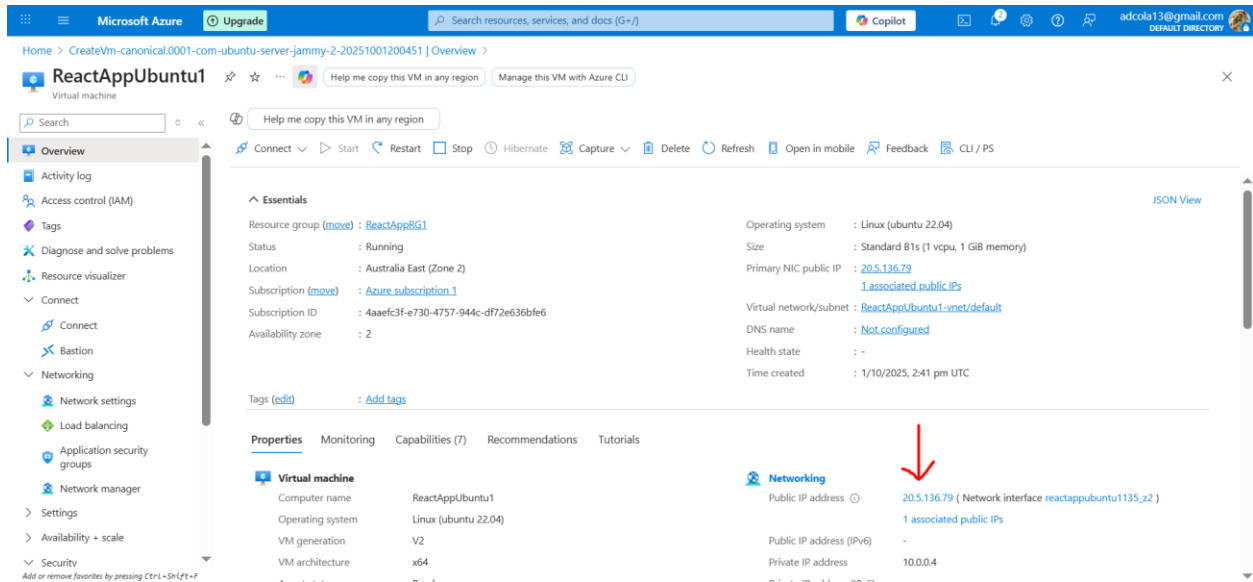
location / {
    try_files $uri /index.html;
}

error_page 404 /index.html;
}' | sudo tee /etc/nginx/sites-available/default > /dev/null
21  sudo systemctl restart nginx
22  curl ifconfig.me
23  history
azureuser@ReactAppUbuntu1:~/my-react-app$
```

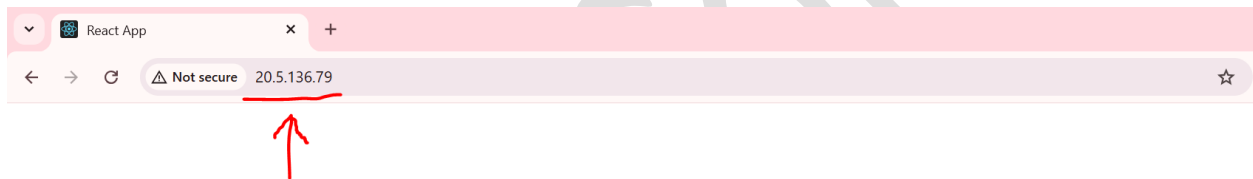
Access it in your browser:

<http://<your-public-ip>>

Public URL of app: <http://20.5.136.79/>



Screenshot of the browser view



Welcome to My React App

This app is running on Nginx!

Deployed by: **Adrina Colaco**

Date: **01/Oct/2025**

P.S. This post is part of the FREE DevOps for Beginners Cohort run by **Pravin Mishra**.

You can start your DevOps journey for free from his [YouTube Playlist](#).

Connect with Pravin on [LinkedIn](#).

Assignment 25 – Design a Three-Tier Network with Load Balancer on Azure

1. Create Resource Group

Microsoft Azure portal interface showing the 'VnetDemoRG' resource group. The left sidebar lists navigation options: Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Cost Management, Monitoring, Automation, and Help. The main content area displays 'Essentials' with 'Resources' and 'Recommendations' tabs. A message states: 'No resources match your filters. Try changing or clearing your filters.' Below this message are buttons for '+ Create' and 'Clear filters', and a link for 'Learn more'.

2. Create a Virtual Network

Microsoft Azure portal interface showing the 'ebDemoVnet | Subnets' page. The left sidebar lists navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Address space, Connected devices, Subnets, Bastion, DDoS protection, Firewall, and Microsoft Defender for Cloud. The main content area displays a table of subnets. A red box highlights the 'Subnets' tab in the left sidebar and the 'Subnets' table in the main content area.

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
default	10.0.0.0/24	-	251	-	-	-
AzureBastionSubnet	10.0.1.0/26	-	57	-	-	-
web-subnet	10.0.2.0/24	-	250	-	-	-
app-subnet	10.0.3.0/25	-	123	-	-	-
db-subnet	10.0.4.0/26	-	59	-	-	-

3. Create a VM in the Web Subnet

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > CreateVm-canonicalubuntu-24_04-lts-server-20251001231719 | Overview >

web-nginx Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search | Help me copy this VM in any region

Overview | Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Open in mobile | Feedback | CLI / PS

Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Resource visualizer | Connect | Networking | Settings | Availability + scale | Security | Backup + disaster recovery | Operations | Monitoring | Automation | Help

Resource group (move) : VnetDemoRG

Status : Running

Location : Australia East (Zone 1)

Subscription (move) : Azure subscription 1

Subscription ID : 4aaefc3f-e730-4757-944c-df72e636bfe6

Availability zone : 1

Tags (edit) : Add tags

Operating system : Linux (ubuntu 24.04)

Size : Standard B1s (1 vcpu, 1 GiB memory)

Primary NIC public IP : 20.5.40.18 | 1 associated public IP

Virtual network/subnet : ebDemoVnet/web-subnet

DNS name : Not configured

Health state : -

Time created : 1/10/2025, 5:52 pm UTC

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Virtual machine

Computer name : web-nginx

Operating system : Linux (ubuntu 24.04)

VM generation : V2

VM architecture : x64

Agent status : Ready

Agent version : 2.150.1

Virtual network

Public IP address : 20.5.40.18 (Network interface web-nginx661_x1) | 1 associated public IP

Public IP address (IPv6) : -

Private IP address : 10.0.2.4

Private IP address (IPv6) : -

Virtual network/subnet : ebDemoVnet/web-subnet

DNS name : Not configured

4. Install and Configure NGINX

ssh -i web-nginx_key.pem azureuser@20.5.40.18

sudo apt update

sudo apt install -y nginx

sudo systemctl enable nginx

sudo systemctl start nginx

```

azureuser@web-nginx:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-01 18:05:18 UTC; 2min 14s ago
     Docs: man:nginx(8)
   Main PID: 2468 (nginx)
    Tasks: 2 (limit: 995)
  Memory: 1.7M (peak: 3.7M)
     CPU: 23ms
   CGroup: /system.slice/nginx.service
           └─2468 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2471 "nginx: worker process"

Oct 01 18:05:18 web-nginx systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 01 18:05:18 web-nginx systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
azureuser@web-nginx:~$
  
```

5. Create a Public Load Balancer

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home >

web-public-elb Load balancer

Analyze alerts for this load balancer | Troubleshoot connectivity issues in this load balancer | How many critical alerts for this load balancer

Search | Move | Delete | Refresh | Give feedback

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Resource visualizer | Settings | Monitoring

Resource group (move) : VnetDemoRG

Location (move) : Australia East

Subscription (move) : Azure subscription 1

Subscription ID : 4aaefc3f-e730-4757-944c-df72e636bfe6

SKU : Standard

Tier : Regional

Tags (edit) : Add tags

Backend pool : web-backend-pool (1 virtual machine)

Load balancing rule : web-http-rule (Tcp/80)

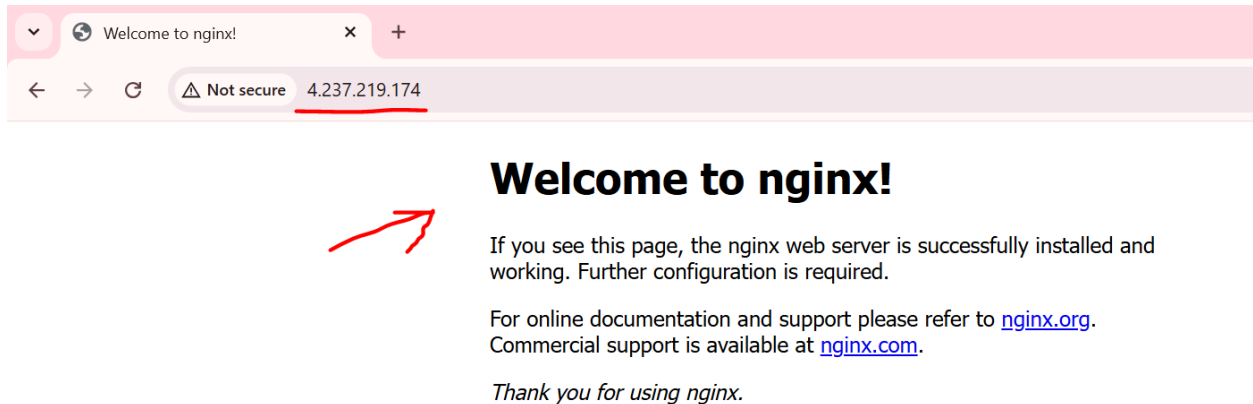
Health probe : web-health-probe (Tcp/80)

Inbound NAT rules : -

Outbound rules : -

Frontend IP address : 4.237.219.174 (LoadBalancer)

6. Test the Architecture



Assignment 26 – Host a Static Website on Azure Storage

1. Clone the repo

```
alanm@DESKTOP-TSHFOQK MINGW64 ~  
$ git clone https://github.com/pravinmishraaws/Azure-Static-Website  
Cloning into 'Azure-Static-Website'...  
remote: Enumerating objects: 7, done.  
remote: Counting objects: 100% (7/7), done.  
remote: Compressing objects: 100% (5/5), done.  
remote: Total 7 (delta 1), reused 7 (delta 1), pack-reused 0 (from 0)  
Receiving objects: 100% (7/7), 4.55 KiB | 518.00 KiB/s, done.  
Resolving deltas: 100% (1/1), done.  
  
alanm@DESKTOP-TSHFOQK MINGW64 ~  
$ cd Azure-Static-Website  
  
alanm@DESKTOP-TSHFOQK MINGW64 ~/Azure-Static-Website (main)  
$ ls  
README.md  index.html
```

2. Personalize

```
MINGW64:/c/Users/alanm/Azure-Static-Website

<div class="cta-row">
  <a class="btn"
    href="https://docs.google.com/forms/d/e/1FAIpQLSexp5B8_Lw3gJI8oc
zG3b7YxHyFbJ-jqX5F58_YBsYa3KMJDQ/viewform"
    target="_blank" aria-label="Pre-register for DMI Cohort 2">Pre-r
egister</a>
  <span class="badge-note">Limited seats</span>
</div>

<footer>
  Hosted By <span id="yr"></span> ADRINA COLACO|.
  <a href="https://www.youtube.com/playlist?list=PLV0dqXbCs7bx88JeUZmK
4fKTq2hJ5VS89" target="_blank"
    rel="noopener">
    Checkout DevOps Micro Internship Cohort 1
  </a>
</footer>
</main>
<script>document.getElementById('yr').textContent = new Date().getFullYear()
</script>
</body>

index.html [dos] (02:05 02/10/2025) 477,59 99%
"index.html" [dos] 487L, 16284B
```

3. Create a Storage Account

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > staticwebsiteadrina_1759351247310 | Overview >

staticwebsiteadrina Storage account

Create a cost-saving LCM rule for this storage account | Analyze alerts for this storage account | Improve data protection for this storage account

Search

Upload | Open in Explorer | Delete | Move | Refresh | Open in mobile | CLI / PS | Feedback

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
- Security + networking
- Data management
- Settings
- Monitoring
- Monitoring (classic)

Essentials

Resource group (move)	: StaticWebsite	Performance	: Standard
Location	: australiaeast	Replication	: Locally-redundant storage (LRS)
Subscription (move)	: Azure subscription 1	Account kind	: StorageV2 (general purpose v2)
Subscription ID	: 4aaefc3f-e730-4757-944c-df72e636bfe6	Provisioning state	: Succeeded
Disk state	: Available	Created	: 10/2/2025, 2:13:06 AM
Tags (edit)	: Add tags		

Properties | Monitoring | Capabilities (7) | Recommendations (0) | Tutorials | Tools + SDKs

Blob service

Hierarchical namespace	: Disabled
Default access tier	: Hot
Blob anonymous access	: Disabled
Blob soft delete	: Enabled (7 days)
Container soft delete	: Enabled (7 days)
Versioning	: Disabled
Change feed	: Disabled
NFS v3	: Disabled

Security

Require secure transfer for REST API operations	: Enabled
Storage account key access	: Enabled
Minimum TLS version	: Version 1.2
Infrastructure encryption	: Disabled

Networking

Public network access	: Enabled
Public network access scope	: Enable from all networks

4. Enable Static Website

Home > staticwebsitedrina_1759351247310 | Overview > staticwebsitedrina

staticwebsitedrina | Static website

Storage account

Search Save Discard Give feedback

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Security + networking
Networking

Enabling static websites on the blob service allows you to host static content. Webpages may include static content and client-side scripts. Server-side scripting is not supported. As data is replicated asynchronously from primary to secondary regions, files at the secondary endpoint may not be immediately available or in sync with files at the primary endpoint. [Learn more](#)

Static website
Disabled Enabled

An Azure Storage container has been created to host your static website.
[\\$web](#)

Improve the page load time of your static website by using the caching features of Azure Front Door (Additional costs apply). [Azure Front Door](#)

Primary endpoint [https://staticwebsitedrina.z8.web.core.windows.net/](#) Copied

Index document name [index.html](#) ✓

Error document path [404.html](#) ✓

5. Upload site files

Home > staticwebsitedrina

staticwebsitedrina | Storage browser

Storage account

Search staticwebsitedrina Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease

Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage

staticwebsitedrina
Favorites
Recently viewed
Blob containers
\$logs
\$web
View all
File shares
Queues
Tables

Blob containers > \$web
Authentication method: Access key (Switch to Microsoft Entra user account)
Add filter
Search blobs by prefix (case-sensitive)
Only show active blobs

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
README.md	10/2/2025, 2:19:55 AM	Hot (Inferred)	Block blob	24 B	Available
index.html	10/2/2025, 2:19:55 AM	Hot (Inferred)	Block blob	15.9 KiB	Available

6. Test

staticwebsitedrina.z8.web.core.windows.net

Mandatory learning & regular reshuffling — no passengers.

- Principle: no shortcuts — doing is how you learn
- Submit **all** assignments on time
- **Show your work** on LinkedIn/blogs
- **Help peers** grow with you

Missing without valid reason → seat given to a more committed learner

Champion of the Week — own outcomes, get recognized.

- From **Week 1**, join a project squad
- Build and ship as a team

Weekly Champion per squad (impact • ownership • collaboration)

Yes, it's still FREE — reserved for committed learners only.

Pre-register • **Limited seats**

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Assignment 27 – Deploy EpicBook on Azure VM + Azure Database for MySQL

1) Network & Security

1.1 Create VNet + Subnets(1 for Public and 1 for MySQL)

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > epic-vnet-1759354376237 | Overview > epic-vnet

epic-vnet | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
default	10.0.0.0/24	-	251	-	-	-
public-subnet	10.0.1.0/24	-	251	-	-	-
mysql-subnet	10.0.2.0/24	-	251	Microsoft.DBforMySQL/flexi...	-	-

1.2 Create Network Security Groups(NSGs)

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > nsg-public

nsg-public | Inbound security rules

Network security group

Search

+ Add Hide default rules Refresh Delete Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	allow-ssh	22	TCP	223.177.210.62	Any	Allow
110	allow-http	80	TCP	0.0.0.0	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerIn...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

1.3 Create a Public IP & NIC to be associated with the Public VM.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > Network foundation | Public IP addresses >

EBpublicIP Public IP address

Can you check if my public ip is highly available | How to ensure public ip fault tolerance

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Associate | Dissociate | Delete | Move | Refresh | Open in mobile | Give feedback

Essentials

Resource group (move) : epicbooks

Location (move) : Australia East

Subscription (move) : Azure subscription 1

Subscription ID : 4aaefc3f-e730-4757-944c-df72e636bfe6

SKU : Standard

Tier : Regional

IP address : 20.227.90.228

DNS name : -

Associated to : -

Virtual machine : -

Tags (edit) : Add tags

Get Started | Properties | Tutorials

Public IP address

Resource GUID : 5bad9d75-17ee-41e0-9904-93da46400bd

IP address : 20.227.90.228

IP address version : IPv4

Provisioning state : Succeeded

Allocation method : Static

Domain name label scope : -

Idle timeout (change) : 4

Associated to : -

DDoS Protection

Protection status : Not connected to backend

Protection type : Network inherited protection

Plan ID : -

2) Provision the Azure VM

Note: The IP address will change after attaching the Public IP to this VM.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/)

Home > CreateVm-canonical.ubuntu-24_04-lts-server-20251002103627 | Overview >

epicbook-vm Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
- Networking
- Network settings
- Load balancing
- Application security groups
- Network manager
- Settings
- Disks
- Extensions + applications
- Operating system

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Open in mobile | Feedback | CLI / PS

Help me copy this VM in any region

Essentials

Resource group (move) : epicbooks

Status : Running

Location : Australia East (Zone 1)

Subscription (move) : Azure subscription 1

Subscription ID : 4aaefc3f-e730-4757-944c-df72e636bfe6

Availability zone : 1

Operating system : Linux (ubuntu 24.04)

Size : Standard B1s (1 vcpu, 1 GiB memory)

Primary NIC public IP : 20.211.149.37

Virtual network/subnet : epic-vnet/public-subnet

DNS name : Not configured

Health state : -

Time created : 2/10/2025, 5:12 am UTC

Tags (edit) : Add tags

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Virtual machine

Computer name : epicbook-vm

Operating system : Linux (ubuntu 24.04)

VM generation : V2

VM architecture : x64

Networking

Public IP address () : 20.211.149.37 (Network interface epicbook-vm580_z1)

Public IP address (IPv6) : -

Private IP address : 10.0.1.4

IP of the VM changed when it was associated with previously created Public IP.

The screenshot shows the 'IP configurations' page for the network interface 'epicbook-vm580_z1'. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Settings, IP configurations (selected), DNS servers, Network security group, Properties, Locks, Monitoring, Automation, and Help. The main content area is titled 'IP Settings' and includes options to 'Enable IP forwarding', 'Virtual network' (set to 'epic-vnet'), and 'Gateway load balancer' (set to 'None'). Below these is a table of IP configurations:

Name	IP Version	Type	Private IP Address	Public IP Address
ipconfig1	IPv4	Primary	10.0.1.4 (Dynamic)	20.227.90.228 (EBpublicIP)

Buttons for '+ Add', 'Make primary', and 'Delete' are visible above the table. A note explains that private and public IP addresses can be assigned to a virtual machine's network interface controller. The bottom of the page has 'Apply' and 'Discard changes' buttons.

The screenshot shows the 'Overview' page for the virtual machine 'epicbook-vm'. The left sidebar includes navigation options like Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Connect, Networking, Network settings, Load balancing, Application security groups, Network manager, Settings, Disks, Extensions + applications, and Operating system. The main content area is titled 'Overview' and includes a 'Help me copy this VM in any region' button. Below this is a table of 'Essentials' information:

Property	Value
Resource group	epicbooks
Status	Running
Location	Australia East (Zone 1)
Subscription	Azure subscription 1
Subscription ID	4aaefc3f-e730-4757-944c-df72e636bfe6
Availability zone	1

Below the essentials table is a 'Tags' section with an 'Add tags' button. The 'Properties' tab is selected, showing details about the 'Virtual machine' (Computer name: epicbook-vm, Operating system: Linux (ubuntu 24.04), VM generation: V2, VM architecture: x64). The 'Networking' tab is also visible, showing the 'Public IP address' (20.227.90.228) and 'Private IP address' (10.0.1.4). A red arrow points to the 'Networking' tab.

ADD PORT 8080 to the NSG of your VM

[Attach network interface](#)
[Detach network interface](#)
[View topology](#)
[Troubleshoot](#)
[Refresh](#)
[Give feedback](#)

Network interface / IP configuration
epicbook-vm580_z1 (primary) / ipconfig1 (primary)

Essentials

Network interface	: epicbook-vm580_z1	Load balancers	: 0 (Configure)
Virtual network / subnet	: epic-vnet / public-subnet	Application security groups	: 0 (Configure)
Public IP address	: 20.227.90.228	Network security group	: epicbook-vm-nsg
Private IP address	: 10.0.1.4	Accelerated networking	: Disabled
Admin security rules	: 0 (Configure)	Effective security rules	: 0

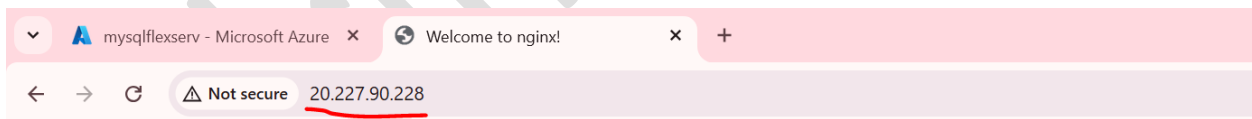
Rules [Collapse all](#)

Network security group **epicbook-vm-nsg** (attached to networkInterface: epicbook-vm580_z1)
 Impacts 0 subnets, 1 network interfaces
 [+ Create port rule](#)

Search rules
 [Source == all](#)
[Destination == all](#)
[Protocol == all](#)
[Action == all](#)
[Port == all](#)

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (7)						
300	SSH	22	TCP	Any	Any	Allow
320	HTTP	80	TCP	Any	Any	Allow
330	MySQL	3306	TCP	223.177.210.62	Any	Allow
340	Http8080	8080	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

Testing the Original NGINX page



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org. Commercial support is available at nginx.com.

Thank you for using nginx.

3) Deploy EpicBook App

3.1 Clone repo into your VM

```

azureuser@epicbook-vm:~$ cd ~
azureuser@epicbook-vm:~$ git clone https://github.com/pravinmishraaws/theepicbook.git
Cloning into 'theepicbook'...
remote: Enumerating objects: 156, done.
remote: Counting objects: 100% (156/156), done.
remote: Compressing objects: 100% (146/146), done.
remote: Total 156 (delta 43), reused 93 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (156/156), 30.70 MiB | 13.27 MiB/s, done.
Resolving deltas: 100% (43/43), done.
azureuser@epicbook-vm:~$ cd theepicbook
azureuser@epicbook-vm:~/theepicbook$ ls
'Installation & Configuration Guide.md'  config  models  package.json  routes  views
README.md                                db      package-lock.json  public  server.js
azureuser@epicbook-vm:~/theepicbook$ █

```

3.2 Install project dependencies into your VM

sudo apt update

sudo apt upgrade -y

sudo apt install nodejs npm

```

azureuser@epicbook-vm:~$ node -v
v18.19.1
azureuser@epicbook-vm:~$ npm -v
9.2.0
azureuser@epicbook-vm:~$ █

```

Install NPM

```

azureuser@epicbook-vm:~/theepicbook$ npm install
npm WARN deprecated ini@1.3.5: Please update to ini >=1.3.6 to avoid a prototype pollution issue
npm WARN deprecated debug@4.1.1: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)
npm WARN deprecated debug@4.1.1: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)
npm WARN deprecated debug@3.2.6: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when used in a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/issues/797)
added 357 packages, and audited 358 packages in 17s

30 packages are looking for funding
  run `npm fund` for details

37 vulnerabilities (5 low, 10 moderate, 18 high, 4 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.

```

4) Create Azure Database for MySQL – Flexible Server (Private)

Microsoft Azure Upgrade Search resources, services, and docs (G+/J) Copilot adcola13@gmail.com DEFAULT DIRECTORY

Home >

mysqlflexserv Azure Database for MySQL flexible server

Summarize the configurations of this Azure Database for MySQL flexible server. List access control settings for this Azure Database for MySQL flexible server. +1

Delete Reset password Restore Restart Upgrade Stop Refresh Feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Learning center
- Resource visualizer
- Settings
- Power Platform
- Security
- Monitoring
- Automation
- Help

Azure Database for MySQL – Live Webinar series: Learn about the latest updates (with demos) and interact directly with product group on the 2nd Wednesday of every month! [Subscribe to our YouTube channel](#) today!

JSON View

Essentials

Subscription (move) : [Azure subscription 1](#)

Subscription ID : 4aaefc3f-e730-4757-944c-df72e636bfe6

Resource group (move) : [epicbooks](#)

Status : Ready

Location : Australia East

Server name : mysqlflexserv.mysql.database.azure.com

Administrator login : azureuser

Configuration : [Burstable_81ms_1 vCores_2 GiB RAM_64 storage_492 IOPS](#)

MySQL version : [8.0 Upgrade](#)

Availability zone : 1

Created on : 2025-10-02 06:25:40.0617701 UTC

Tags (edit) : [Add tags](#)

Getting started Properties Recommendations Monitoring Tutorials

Compute + storage (Change)

Pricing tier : Burstable

Compute size : Standard_B1ms (1 vCore, 2 GiB memory, 640 max iops)

Storage : 64 GiB

IOPS : Auto scale IOPS

Accelerated Logs : Disabled

Storage autogrow : Enabled

Networking (Change)

Connectivity method : Virtual network

Private DNS zone : Private DNS zone

Private access (VNet Integration) : [epic-vnet/mysql-subnet](#)

[mysqlflexserv.private.mysql.database.azure.com](#)

High availability (Change)

High availability : Not enabled

Add or remove favorites by pressing Ctrl+Shift+F

```
azureuser@epicbook-vm:~$ sudo mysql -h mysqlflexserv.mysql.database.azure.com -u azureuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.42-azure Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

4.1 Install MYSQL on the Azure VM.

```
azureuser@epicbook-vm:~/theepicbook$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@epicbook-vm:~/theepicbook$
azureuser@epicbook-vm:~/theepicbook$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-template-perl libmecab2
  libprotobuf-lite32t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl mailx tinycat
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-template-perl libmecab2
  libprotobuf-lite32t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 15 newly installed, 0 to remove and 0 not upgraded.
Need to get 26.5 MB of archives.
After this operation, 180 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [7982 B]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libmecab2 amd64 0.996-14ubuntu4 [201 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-8.2ubuntu0.2 [238 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-core-8.0 amd64 8.0.43-0ubuntu0.24.04.2 [17.5 MB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-server-8.0 amd64 8.0.43-0ubuntu0.24.04.2 [1439 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libcgi-pm-perl all 4.63-1 [185 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfcgi0t64 amd64 2.4.2-2.1ubuntu0.24.04.1 [27.0 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libfcgi-perl amd64 0.82+ds-3build2 [21.7 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libcgi-fast-perl all 1:2.17-1 [10.3 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfcgi-bin amd64 2.4.2-2.1ubuntu0.24.04.1 [11.2 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libhtml-template-perl all 2.97-2 [60.2 kB]
mysql-server is already the newest version (8.0.43-0ubuntu0.24.04.2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@epicbook-vm:~$ sudo systemctl enable mysql
sudo systemctl start mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql

azureuser@epicbook-vm:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-02 06:51:47 UTC; 9h ago
   Process: 10591 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
   Main PID: 10600 (mysqld)
   Status: "Server is operational"
     Tasks: 37 (limit: 993)
    Memory: 405.9M (peak: 409.9M)
       CPU: 2min 44.307s
    CGroup: /system.slice/mysql.service
           └─10600 /usr/sbin/mysqld

Oct 02 06:51:46 epicbook-vm systemd[1]: Starting mysql.service - MySQL Community Server...
Oct 02 06:51:47 epicbook-vm systemd[1]: Started mysql.service - MySQL Community Server.

azureuser@epicbook-vm:~$ ping mysqlflexserv.mysql.database.azure.com
PING mysqlflexserv.mysqlflexserv.private.mysql.database.azure.com (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=63 time=2.82 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=63 time=1.29 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=63 time=1.36 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=63 time=1.04 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=63 time=1.27 ms
```

Log into MYSQL from within AZURE VM.

```
azureuser@epicbook-vm:~$ sudo mysql -h mysqlflexserv.mysql.database.azure.com -u azureuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.42-azure Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

4.2 Create DB & import schema

```
mysql> CREATE DATABASE bookstore;
Query OK, 1 row affected (0.06 sec)
```

Switch to bookstore database and create the needed tables.

```
mysql> USE bookstore;
Database changed
mysql> CREATE TABLE `bookstore`.`Author` (
  ->   `id` INT NOT NULL AUTO_INCREMENT,
  ->   `firstName` VARCHAR(45) NOT NULL,
  ->   `lastName` VARCHAR(45) NOT NULL,
  ->   `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ->   PRIMARY KEY (`id`));
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> CREATE TABLE `bookstore`.`Book` (  
-> `id` INT NOT NULL AUTO_INCREMENT,  
-> `title` VARCHAR(255) NOT NULL,  
-> `genre` VARCHAR(255) NOT NULL,  
-> `pubYear` INT NOT NULL,  
-> `price` DECIMAL(13,2) NOT NULL,  
-> `inventory` INT NOT NULL,  
-> `bookDescription` TEXT NOT NULL,  
-> `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
-> `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
-> `AuthorId` INT NOT NULL,  
-> PRIMARY KEY (`id`),  
-> INDEX `AuthorId_idx` (`AuthorId` ASC),  
-> CONSTRAINT `AuthorId`  
-> FOREIGN KEY (`AuthorId`)  
-> REFERENCES `bookstore`.`Author` (`id`)  
-> ON DELETE NO ACTION  
-> ON UPDATE NO ACTION);  
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> CREATE TABLE `bookstore`.`Cart` (  
-> `id` INT NOT NULL AUTO_INCREMENT,  
-> `quantity` INT NOT NULL,  
-> `price` DECIMAL(13,2) NOT NULL,  
-> `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
-> `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
-> PRIMARY KEY (`id`));  
Query OK, 0 rows affected (0.07 sec)
```

Insert records into the tables

```
mysql> INSERT INTO Author(firstName,lastName,createdAt,updatedAt)
-> VALUES
-> ('Elin','Hildebrand',sysdate(),sysdate())
-> ,('Brit','Bennett',sysdate(),sysdate())
-> ,('John','Bolton',sysdate(),sysdate())
-> ,('Robin','DiAngelo',sysdate(),sysdate())
-> ,('Delia','Owens',sysdate(),sysdate())
-> ,('Hortense','Spillers',sysdate(),sysdate())
,('Stephen','Best',sysdate(),sysdate())
,('Anthony B.','Pinn',sysdate(),sysdate())
,('Harvey','Young',sysdate(),sysdate())
,('Michael A.','Gomez',sysdate(),sysdate())
,('Sarah','Haley',sysdate(),sysdate())
,('Daryl Michael','Scott',sysdate(),sysdate())
,('Cheryl','Harris',sysdate(),sysdate())
,('Frank','Herbert',sysdate(),sysdate())
,('Andrzej','Sapkowski',sysdate(),sysdate())
,('J.R.R.','Tolkein',sysdate(),sysdate())
,('J.K.','Rowling',sysdate(),sysdate())
,('Stephenie','Meyer',sysdate(),sysdate())
,('George R. R.','Martin',sysdate(),sysdate())
,('Diana','Gabaldon',sysdate(),sysdate())
,('Robert','Jordan',sysdate(),sysdate())
,('Cixin','Liu',sysdate(),sysdate()),('John','Grisham',sysdate(),sysdate())
-> ,('Ibram X.','Kendi',sysdate(),sysdate())
-> ,('Celeste','Ng',sysdate(),sysdate())
-> ,('Ijeoma','Oluo',sysdate(),sysdate())
-> ,('Harper','Lee',sysdate(),sysdate())
-> ,('L.M.','Montgomery',sysdate(),sysdate())
-> ,('Jack','London',sysdate(),sysdate())
-> ,('Jane','Austen',sysdate(),sysdate())
-> ,('F. Scott','Fitzgerald',sysdate(),sysdate())
-> ,('Emily','Bronte',sysdate(),sysdate())
-> ,('Oscar','Wilde',sysdate(),sysdate())
-> ,('George','Orwell',sysdate(),sysdate())
-> ,('J.D.','Salinger',sysdate(),sysdate())
-> ,('Shel','Silverstein',sysdate(),sysdate())
-> ,('Maurice','Sendak',sysdate(),sysdate())
-> ,('Dr.','Suess',sysdate(),sysdate())
-> ,('Beatrix','Potter',sysdate(),sysdate())
-> ,('C.S.','Lewis',sysdate(),sysdate())
-> ,('E.B.','White',sysdate(),sysdate())
-> ,('Eric','Carle',sysdate(),sysdate())
-> ,('Michael','Ende',sysdate(),sysdate())
-> ,('Howard','Zinn',sysdate(),sysdate())
-> ,('Norton','Juster',sysdate(),sysdate())
-> ,('Kurt','Vonnegut',sysdate(),sysdate())
-> ,('Chinua','Achebe',sysdate(),sysdate())
-> ,('Kazuo','Ishiguro',sysdate(),sysdate())
```

```
-> ,('Art','Spiegelman',sysdate(),sysdate())
-> ,('Margaret','Atwood',sysdate(),sysdate())
-> ,('Douglas','Adams',sysdate(),sysdate())
-> ,('David Foster','Wallace',sysdate(),sysdate())
-> ,('Philip D.','Morgan',sysdate(),sysdate())
-> ,('Hortense','Spillers',sysdate(),sysdate())
-> ,('Stephen','Best',sysdate(),sysdate())
-> ,('Anthony B.','Pinn',sysdate(),sysdate())
-> ,('Harvey','Young',sysdate(),sysdate())
-> ,('Michael A.','Gomez',sysdate(),sysdate())
-> ,('Sarah','Haley',sysdate(),sysdate())
-> ,('Daryl Michael','Scott',sysdate(),sysdate())
-> ,('Cheryl','Harris',sysdate(),sysdate())
-> ,('Frank','Herbert',sysdate(),sysdate())
-> ,('Andrzej','Sapkowski',sysdate(),sysdate())
-> ,('J.R.R.','Tolkien',sysdate(),sysdate())
-> ,('J.K.','Rowling',sysdate(),sysdate())
-> ,('Stephenie','Meyer',sysdate(),sysdate())
-> ,('George R. R.','Martin',sysdate(),sysdate())
-> ,('Diana','Gabaldon',sysdate(),sysdate())
-> ,('Robert','Jordan',sysdate(),sysdate())
-> ,('Cixin','Liu',sysdate(),sysdate());
Query OK, 53 rows affected (0.03 sec)
Records: 53 Duplicates: 0 Warnings: 0

mysql> █
```

```

mysql> INSERT INTO Book(title,genre,pubYear,price,inventory,bookDescription,createdAt,updatedAt,AuthorId)
-> VALUES
-> ('28 Summers','NTT','2020','28','4','A relationship that started in 1993 between Mallory Blessing and Jake McCloud comes to light while she is on her deathbed and his wife runs for presi
dent.',sysdate(),sysdate(),1)
-> ,('The Vanishing Half','NTT','2020','27','8','The lives of twin sisters who run away from a Southern Black community at age 16 diverge as one returns and the other takes on a different ra
cial identity but their fates intertwine.',sysdate(),sysdate(),2)
-> ,('The Room Where it Happened','NTT','2020','32.50','7','The former national security advisor gives his account of the 17 months he spent working for President Trump.',sysdate(),sysdate(),
3)
-> ,('White Fragility','NTT','2018','16','4','Historical and cultural analyses on what causes defensive moves by white people and how this inhibits cross-racial dialogue.',sysdate(),sysdate(),
4)
-> ,('Where the Crawdads Sing','NTT','2018','26','7','In a quiet town on the North Carolina coast in 1969, a young woman who survived alone in the marsh becomes a murder suspect.',sysdate(),
sysdate(),5)
-> ,('Camino Winds','NTT','2020','28.95','9','The line between fact and fiction becomes blurred when an author of thrillers is found dead after a hurricane hits Camino Island.',sysdate(),sys
date(),6)
-> ,('How to be an Antiracist','NTT','2019','27','3','A primer for creating a more just and equitable society through identifying and opposing racism.',sysdate(),sysdate(),7)
-> ,('Little Fires Everywhere','NTT','2017','17','5','An artist with a mysterious past and a disregard for the status quo upends a quiet town outside Cleveland.',sysdate(),sysdate(),8)
-> ,('So You Want to Talk About Race','NTT','2019','16.99','2','A look at the contemporary racial landscape of the United States.',sysdate(),sysdate(),9)
-> ,('To Kill a Mockingbird','Classic','1960','15.99','10','In "To Kill a Mockingbird," author Harper Lee uses memorable characters to explore civil rights and racism in the segregated South
ern United States of the 1930s. Told through the eyes of Scout Finch, you learn about her father Atticus Finch, an attorney who hopelessly strives to prove the innocence of a black man unjustly
accused of rape; and about Boo Radley, a mysterious neighbor who saves Scout and her brother Jem from being killed.',sysdate(),sysdate(),10)
-> ,('Anne of Green Gables','Classic','1908','5.99','8','Anne of Green Gables is the story of a young orphan girl, Anne, who is mistakenly delivered to an older couple looking to adopt a boy
to work on their farm in Avonlea on Prince Edward Island. After the couple meets Anne, they decide to keep her despite her not being of the appropriate gender.',sysdate(),sysdate(),11)
-> ,('The Call of the Wild','Classic','1903','12.95','6','Based on Jack London's experiences as a gold prospector in the Canadian wilderness and his ideas about nature and the struggle for
existence, The Call of the Wild is a tale about unbreakable spirit and the fight for survival in the frozen Alaskan Klondike.',sysdate(),sysdate(),12)
-> ,('Pride and Prejudice','Classic','1913','9.00','9','A classic of English literature, written with incisive wit and superb character delineation, it centres on the turbulent relationship
between Elizabeth Bennet, the daughter of a country gentleman, and Fitzwilliam Darcy, a rich aristocratic landowner.',sysdate(),sysdate(),13)
-> ,('The Great Gatsby','Classic','1925','17.00','4','The story is of the fabulously wealthy Jay Gatsby and his new love for the beautiful Daisy Buchanan, of lavish parties on Long Island at
a time when The New York Times noted "gin was the national drink and sex the national obsession," it is an exquisitely crafted tale of America in the 1920s.',sysdate(),sysdate(),14)
-> ,('Wuthering Heights','Classic','1847','8.90','7','Wuthering Heights is a classic gothic horror/romance novel written by Emily Brontë and published in England in 1847. The book was original
ly published under the pen name Ellis Bell and Brontë died the year after it was released. When the book was first released it was considered controversial for its themes of violence and it s
challenge of Victorian ideals.',sysdate(),sysdate(),15)
-> ,('The Picture of Dorian Gray','Classic','1890','6.99','5','In the stately London home of his aunt, Lady Brandon, the well-known artist Basil Hallward meets Dorian Gray. Dorian is a cultu
red, wealthy, and impossibly beautiful young man who immediately captures Basil's artistic imagination. Dorian sits for several portraits, and Basil often depicts him as an ancient Greek her
o or a mythological figure. The Picture of Dorian Gray is the story of one beautiful, innocent young man's seduction, moral corruption, and eventual downfall.',sysdate(),sysdate(),16)
-> ,('Animal Farm','Classic','1945','18.50','3','One of Orwell's finest works, it is a political fable based on the events of Russia's Bolshevik revolution and the betrayal of the cause by
Joseph Stalin. The book concerns a group of barnyard animals who overthrow and chase off their exploitative human masters and set up an egalitarian society of their own.',sysdate(),sysdate(),
17)
-> ,('The Catcher in the Rye','Classic','1951','9.99','2','The novel details two days in the life of 16-year-old Holden Caulfield after he has been expelled from prep school. Confused and dis
illusioned, Holden searches for truth and rails against the phoniness of the adult world. He ends up exhausted and emotionally unstable.',sysdate(),sysdate(),18)
-> ,('The Giving Tree','Childrens','1964','17.99','8','The book follows the lives of an apple tree and a boy, who develop a relationship with one another. The tree is very "giving" and the b
oy evolves into a "taking" teenager, man, then elderly man. Despite the fact that the boy ages in the story, the tree addresses the boy as "Boy" his entire life.',sysdate(),sysdate(),24)
-> ,('Where the Wild Things Are','Childrens','1963','8.95','9','This story focuses on a young boy named Max who, after dressing in his wolf costume, wreaks such havoc through his household t
hat he is sent to bed without his supper. Max's bedroom undergoes a mysterious transformation into a jungle environment, and he winds up sailing to an island inhabited by malicious beasts k
nown as the "Wild Things".',sysdate(),sysdate(),19)
-> ,('The Cat in the Hat','Childrens','1957','9.99','4','The Cat in the Hat is a book where an eccentric stranger (who's a cat!) comes into the house of two young children,Sally and Sam, wh
o are having a very dull day. Their mother is out, and when the Cat comes in, he reassures the kids that their mother won't mind him or his tricks!.',sysdate(),sysdate(),20)
-> ,('The Tale of Peter Rabbit','Childrens','1902','7.99','6','The widowed mother rabbit warns her four rabbit children, Flopsy, Mopsy, Cottontail and Peter not to enter the vegetable garden
of a man named Mr. McGregor, whose wife, she tells them, put their father in a pie after he entered. Her triplets (Flopsy, Mopsy and Cottontail) obediently refrain from entering the garden,
but Peter enters the garden to snack on some vegetables.',sysdate(),sysdate(),21)
-> ,('The Lion, The Witch, and The Wardrobe','Childrens','1950','8.99','5','Peter, Susan, Edmund, and Lucy Pevensie are four siblings sent to live in the country with the eccentric Professor
Kirke during World War II. The children explore the house on a rainy day and Lucy, the youngest, finds an enormous wardrobe. Lucy steps inside and finds herself in a strange, snowy wood.',s
ysdate(),sysdate(),22)
-> ,('Charlotte's Web','Childrens','1952','9.99','9','A masterpiece of American childrens literature, Charlotte's Web is a fable by E.B. White about a runt of a pig named Wilbur, who is lo
ved by a little girl and befriended by a very clever spider named Charlotte.',sysdate(),sysdate(),23)
-> ,('Where the Sidewalk Ends','Childrens','1974','19.99','7','Where the Sidewalk Ends is a collection of quirky and imaginative poems by Shel Silverstein. The poems are accompanied by illus
trations that capture the often silly, peculiar nature of Silverstein's witty words.',sysdate(),sysdate(),24)
-> ,('The Very Hungry Caterpillar','Childrens','1969','5.99','5','This story follows the life of a growing caterpillar who begins as an egg. When he emerges from the egg he is very hungry. He
starts to eat everything he can find, more and more each day.',sysdate(),sysdate(),25)
-> ,('The Neverending Story','Childrens','1979','8.99','3','The book centres on a boy, Bastian, a strange child who is neglected by his father after the death of Bastian's mother. While esc
aping from some bullies, Bastian bursts into the antiquarian book store of Carl Conrad Coreander, where he finds his interest held by a book called The Neverending Story.',sysdate(),sysdate(),
26)
-> ,('A People's History of the United States','Top 9','1980','21.99','5','A People's History of the United States is the only volume to tell America's story from the point of view of and
in the words of America's women, factory workers, African-Americans, Native Americans, the working poor, and immigrant laborers.',sysdate(),sysdate(),27)
-> ,('The Phantom Tollbooth','Top 9','1961','8.99','6','The Phantom Tollbooth is the story of Milo, a very bored boy who comes home one day to find a magical tollbooth in his room. When Milo
drives his car through the tollbooth gate, he finds himself in the Lands Beyond, a country inhabited by living language in the forms of animals, magicians, royalty, mountains, seas, and cit
ies. From Lock the Watchdog to the listless region of The Doldrums, Milo shakes off boredom as he pursues the kidnapped Princesses Rhyme and Reason and restores peace to the Lands (currently
in the clutches of the warring princes, Azaz of Dictionopolis and the Mathematician of Digitopolis, along with a pack of demons).',sysdate(),sysdate(),28)
-> ,('Slaughterhouse-Five','Top 9','1969','17.00','10','Slaughterhouse-Five, an American classic, is one of the worlds great antiwar books. Centering on the infamous firebombing of Dresden,
Billy Pilgrim's odyssey through time reflects the mythic journey of our own fractured lives as we search for meaning in what we fear most.',sysdate(),sysdate(),29)
-> ,('Things Fall Apart','Top 9','1958','13.00','11','It is a classic narrative about Africa's cataclysmic encounter with Europe as it establishes a colonial presence on the continent. Told
through the fictional experiences of Okonkwo, a wealthy and fearless Igbo warrior of Umuofia in the late 1800s, Things Fall Apart explores one man's futile resistance to the devaluing of h
is Igbo traditions by British political and religious forces and his despair as his community capitulates to the powerful new order.',sysdate(),sysdate(),30)
-> ,('Never Let Me Go','Top 9','2005','16.00','4','Friends Kathy, Tommy and Ruth grow up together at a seemingly idyllic boarding school in the English countryside. When they leave the schoo
l and the horrible truth of their true purpose is revealed to them, they must simultaneously confront deep-seated feelings of love, jealousy and betrayal that threaten to tear their friendsh
ip asunder.',sysdate(),sysdate(),31)
-> ,('Maus','Top 9','1991','35.00','8','Maus depicts Spiegelman interviewing his father about his experiences as a Polish Jew and Holocaust survivor. The work employs postmodernist technique
s and represents Jews as mice, Germans as cats, and Poles as pigs.',sysdate(),sysdate(),32)
-> ,('The Handmaid's Tale','Top 9','1985','15.95','7','The world of the narrator, Offred (from "Of Fred" women no longer have their own names), is chilling, but she is a magnificent survivo
r and chronicler, and the details of everything from mundane daily life to ritualized sex and violence to her reminiscences of the time before (our contemporary reality, as seen in the '80s
) are absolutely realistic.',sysdate(),sysdate(),33)
-> ,('The Hitchhiker's Guide to the Galaxy','Top 9','1980','16.00','8','A hapless hero with astonishing luck? Ill-tempered aliens hell-bent on destroying Earth? Pitthy advice (e.g., "DON'T
PANIC")? Check, check, and check and so much more. Even nonsci-fi geeks will be charmed by this hilarious and endlessly entertaining read, with (of course) sequels following.',sysdate(),sysd
ate(),34)
-> ,('Infinite Jest','Top 9','1996','20.00','9','Infinite Jest is unique; it was bred in the optimism and new frontierism of the dot-com 1990s but was simultaneously an early omen of wh
ere we are today. It looks into our present beyond what were only horizons when it was written: the tensions of a global economy, the opiate of on-demand entertainment, the near-impossible p
ursuit of greatness in a winner-take-all society. Tennis phenoms struggle in an absurdly demanding academy and recovering addicts search for something strong enough to help them through, all
while a cadre of legless Quebecois assassins search for a movie so entertaining that they plan to use it as a weapon.',sysdate(),sysdate(),35)
-> ,('Origins of American Slavery','Social Justice','2005','17.00','3','A concise but thorough overview of how American slavery fits into larger historical processes of the subjugation of bl
ack people, Philip Morgan's description of slavery as an international institution looks at the institutions centrality in shaping global trends throughout the eighteenth and nineteenth centu
ries. By placing American slavery in a global context, Morgan explains how the expanding scale of capitalism and a long-standing perception of black inferiority converged to produce an insta
ntiation of slavery that stands out as peculiarly heinous.',sysdate(),sysdate(),36)
-> ,('Mama's Baby, Papa's Maybe: An American Grammar Book','Social Justice','1987','13.00','5','Hortense Spillers argues that discourses around blackness and gender during slavery continue
to determine the ways in which black bodies are read in the United States. By focusing on black women, Spillers contends that we get it wrong when we only view slavery through the experienc
e of enslaved people who identified as male. The machinations of enslavement as an institution fundamentally relied on fully disarticulating black women from the categories of woman and huma
n.',sysdate(),sysdate(),37)
-> ,('Neither Lost nor Found: Slavery and the Visual Archive','Social Justice','2011','17.00','7','Stephen Best argues that black people have often been understood as object rather than subj
ects. Best cites a case in which a photograph of two young black boys from the nineteenth century was advertised as rare at an estate sale. It was later discovered that copies of the photogr
aph could be found at the Schomburg Center for Research in Black Culture and on eBay. Best addresses the paradox of the archival turn in studies of slavery: How can we claim to know that whi
ch cannot be known? That is, whether evidence can be found in the archive or not?',sysdate(),sysdate(),38)

```

```

-> ,( 'Black Bodies in Pain and Ecstasy: Terror, Subjectivity, and the Nature of Black Religion', 'Social Justice', 2003, '13.00', 4, 'Anthony Pinn argues that black religion, a capacious term that he purposefully deploys to reference a range of religious and spiritual practices beyond Christianity, plays a key role in African Americans struggle for what he calls complex subjectivity, a mode of being defined by ambiguity and multidimensionality. If dominant society defined black people by their corporeality during slavery, Pinn argues that black religion, which is a n experience mediated by and through the black body, constitutes an important site of resistance against oppression. It contains an aesthetic, both in performance and style, that must be more prominently considered by the field of black religious studies.', sysdate(), sysdate(), 39)
-> ,( 'The Black Body as Souvenir in American Lynching', 'Social Justice', 2005, '16.00', 3, 'Lynching, a phenomenon of extra-judicial violence used as a tool of social control, continues to be a lesser acknowledged practice in American history. Harvey Young analyzes what it meant for white participants in lynching spectacles to either steal or purchase the body parts of those who had been unjustly hanged, burned, castrated, and otherwise victimized and killed in public. Young asks what it means for white people to treat black bodies as souvenirs, fetish object, and d remains.', sysdate(), sysdate(), 40)
-> ,( 'Of Du Bois and Diaspora: The Challenge of African American Studies', 'Social Justice', 2004, '35.00', 8, 'Michael Gomez focuses on double consciousness, W. E. B. Du Bois term for the experience of African Americans who must simultaneously identify with blackness and Americanness. The nation signified in the latter term oppresses people based on the former term. Gomez makes a case for black intellectuals in the field of African American studies to more thoughtfully engage a larger diasporic approach with their work. Because the study of black people in America is a diasporic project about dispersal, loss, and community building, Gomez's call for African American studies to take the diasporic turn seriously continues to influence the field.', sysdate(), sysdate(), 41)
-> ,( 'Like I Was a Man: Chain Gangs, Gender, and Domestic Carceral Sphere in Jim Crow Georgia', 'Social Justice', 2013, '15.95', 9, 'Sarah Haley argues that the prison industrial complex has been an important site in which racialized conceptions of gender have been consolidated. Focusing on a comprehensive and historic prison reform act passed in Georgia in 1908 that forced imprisoned black women onto chain gangs and introduced a system of parole that compelled black women released from prison to become domestic servants in white homes, Haley argues that these reforms illustrate how black women were stripped of their gender. Incarcerated black women were obligated to perform both domestic labor that was gendered as female and hard physical labor that was gendered as male, but also, as a result of this un-gendering, they were legislated out of the category of the human.', sysdate(), sysdate(), 42)
-> ,( 'How Black Nationalism Became Sui Generis', 'Social Justice', 2012, '16.00', 2, 'Daryl Michael Scott's exploration of the vicissitudes of black nationalism that developed over the course of the early and mid-twentieth century negotiates a tension between black nationalism and other forms of nationalism. While nationalism on its own may be problematic, black nationalism has described everything from black separatism and sovereignty to an imperium in imperio in which black people could self-determine without founding a new nation to a generic notion of racial solidarity among black people in a Pan-African context. The idea continues to have critical purchase in academia.', sysdate(), sysdate(), 43)
-> ,( 'Whiteness as Property', 'Social Justice', 1993, '20.00', 3, 'Cheryl Harris explores the extent to which whiteness became a form of property that had to be protected by juridical and legislative means. As she argues, whiteness became valuable when white people could not be reduced to property under slavery. Given this, it is not only the cultural legacy of enslavement that has kept black people oppressed in the United States, but also a legal system that has continued to treat whiteness as the norm. That system actively excludes people of color from the purview of equal rights and protections while also affording economic benefits exclusively or disproportionately to white Americans.', sysdate(), sysdate(), 44)
-> ,( 'Dune', 'Fantasy', 1965, '49.95', 8, 'Set on the desert planet Arrakis, Dune is the story of the boy Paul Atreides, heir to a noble family tasked with ruling an inhospitable world where the only thing of value is the spice melange, a drug capable of extending life and enhancing consciousness. Coveted across the known universe, melange is a prize worth killing for....', sysdate(), sysdate(), 45)
-> ,( 'The Witcher', 'Fantasy', 1994, '39.99', 8, 'The Witcher is a fantasy series of novels and short stories written by Polish author Andrzej Sapkowski. The series revolves around the titular "witcher", Gertalt of Rivia. In Sapkowski's works, "witchers" are beast hunters who develop supernatural abilities at a young age to battle wild beasts and monsters.', sysdate(), sysdate(), 46)
-> ,( 'Lord of the Rings', 'Fantasy', 1937, '35.95', 8, 'At 33, the age of adulthood among hobbits, Frodo Baggins receives a magic Ring of Invisibility from his uncle Bilbo. Frodo, a Christlike figure, learns that the ring has the power to control the entire world and, he discovers, to corrupt its owner. A fellowship of hobbits, elves, dwarfs, and men is formed to destroy the ring by casting it into the volcanic fires of the Crack of Doom, where it was forged. They are opposed on their harrowing mission by the evil Sauron and his Black Riders. The Lord of the Rings, together with The Hobbit, is considered by many to be the start of the genre known as high fantasy, and these works have had an enormous influence on that genre as a whole.', sysdate(), sysdate(), 47)
-> ,( 'Harry Potter', 'Fantasy', 1999, '55.99', 8, 'Harry Potter, a boy who learns on his eleventh birthday that he is the orphaned son of two powerful wizards and possesses unique magical powers of his own. He is summoned from his life as an unwanted child to become a student at Hogwarts, an English boarding school for wizards. There, he meets several friends who become his closest allies and help him discover the truth about his parents' mysterious deaths.', sysdate(), sysdate(), 48)
-> ,( 'Twilight', 'Fantasy', 2005, '45.00', 8, 'The Twilight saga is a story about love. And vampires. And family. And abstinence. And racism. And the founding of the Mormon faith. And orphans , in a really weird way.', sysdate(), sysdate(), 49)
-> ,( 'A Song of Ice and Fire', 'Fantasy', 1991, '50.00', 8, 'A Song of Ice and Fire is a series of epic fantasy novels by the American novelist and screenwriter George R. R. Martin. He began the first volume of the series, A Game of Thrones, in 1991, and it was published in 1996.', sysdate(), sysdate(), 50)
-> ,( 'Outlander', 'Fantasy', 1991, '70.00', 8, 'Outlander is a series of historical romance science fiction novels by American author Diana Gabaldon. Gabaldon began the first volume of the series, Outlander, in the late 1980s, and it was published in 1991. She has published eight out of a planned ten volumes.', sysdate(), sysdate(), 51)
-> ,( 'Wheel of Time', 'Fantasy', 1984, '32.00', 8, 'Written by Robert Jordan and completed posthumously by Brandon Sanderson, The Wheel of Time is one of the greatest epics of fantasy and a #1 internationally bestselling series. Taking place both in our past and our future, the saga tells the story of a man destined to face the Evil One and save the world or destroy it in the process.', sysdate(), sysdate(), 52)
-> ,( 'Three-Body Problem Trilogy', 'Fantasy', 2008, '50.00', 8, 'From Hugo Award-winning author Cixin Lu, the most prolific and popular science fiction writer in the People's Republic of China, the Remembrance of Earth's Past series is an action-packed, thought-provoking saga of how humanity's first contact with extraterrestrial life divided Earth between those welcoming the superior beings and those fighting the alien invasion.', sysdate(), sysdate(), 53);
Query OK, 54 rows affected (0.04 sec)
Records: 54 Duplicates: 0 Warnings: 0
mysql>

```

```

mysql> SHOW TABLES;
+-----+
| Tables_in_bookstore |
+-----+
| author               |
| book                 |
| cart                 |
+-----+
3 rows in set (0.00 sec)

mysql>

```

5) Configure Nginx (static SPA + API reverse proxy)

5.1 Serve the React build

```
sudo nano /etc/nginx/sites-available/default
```

```

server {
    listen 80 default_server;
    listen [::]:80 default_server;
    server_name _;

    # Frontend (React build)
    root /home/azureuser/theepicbook/build;
    index index.html;

    location / {
        try_files $uri /index.html;
    }

    # Backend API proxy (change 3001 if your backend uses a different port)
    location /api/ {
        proxy_pass http://127.0.0.1:3306/;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header X-Forwarded-For $remote_addr;
        proxy_set_header Connection "";
    }
}

```

Add database details into the config.json file

```

azureuser@epicbook-vm:~/theepicbook$ vim config/config.json

```

```

"development": {
  "username": "azureuser",
  "password": "XXXXXXXXXX",
  "database": "bookstore",
  "host": "mysqlflexserv.mysql.database.azure.com",
  "dialect": "mysql",
  "dialectOptions": {
    "ssl": {
      "rejectUnauthorized": true
    }
  }
}

```

```

azureuser@epicbook-vm:~/theepicbook$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
azureuser@epicbook-vm:~/theepicbook$ sudo systemctl restart nginx
azureuser@epicbook-vm:~/theepicbook$

```

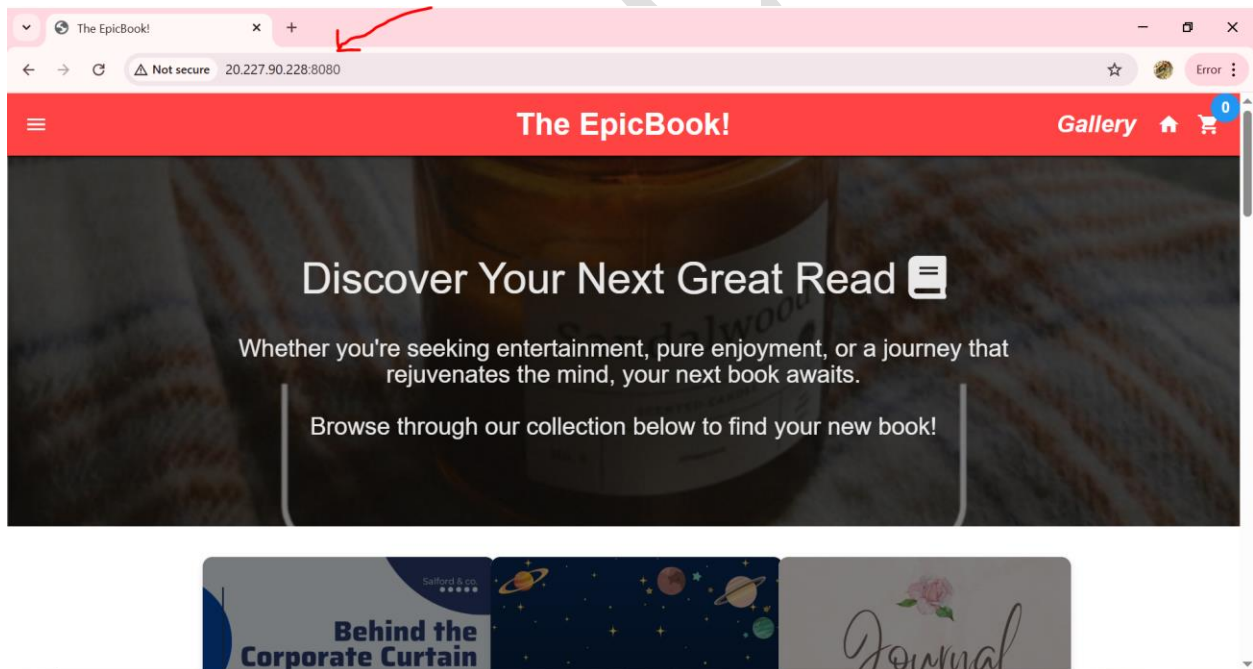
```

azureuser@epicbook-vm:~/theepicbook$ node server.js
going to html route
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Author'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM `Author` FROM `bookstore`
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Book'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM `Book` FROM `bookstore`
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Cart'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM `Cart` FROM `bookstore`
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Checkout'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): CREATE TABLE IF NOT EXISTS `Checkout` (`id` INTEGER NOT NULL auto_increment, `addressLine1` VARCHAR(255)
NOT NULL, `addressLine2` VARCHAR(255) NOT NULL, `city` VARCHAR(255) NOT NULL, `state` VARCHAR(255) NOT NULL, `zipCode` VARCHAR(
255) NOT NULL, `subTotal` DECIMAL NOT NULL, `createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, `CartId` INTEGER,
PRIMARY KEY (`id`), FOREIGN KEY (`CartId`) REFERENCES `Cart` (`id`) ON DELETE SET NULL ON UPDATE CASCADE) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Checkout` FROM `bookstore`
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Cartbook'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): CREATE TABLE IF NOT EXISTS `Cartbook` (`createdAt` DATETIME NOT NULL, `updatedAt` DATETIME NOT NULL, `Book
Id` INTEGER, `CartId` INTEGER, PRIMARY KEY (`BookId`, `CartId`), FOREIGN KEY (`BookId`) REFERENCES `Book` (`id`) ON DELETE
CASCADE ON UPDATE CASCADE, FOREIGN KEY (`CartId`) REFERENCES `Cart` (`id`) ON DELETE CASCADE ON UPDATE CASCADE) ENGINE=InnoDB;
Executing (default): SHOW INDEX FROM `Cartbook` FROM `bookstore`
App listening on PORT 8080

```

6) End-to-End Test

Public IP of the VM with port 8080.



Video of EpicBooks Website Showing books being added to cart and order being placed



EpicBooks_Azure.mp

Request Sent to the Server

```
azureuser@epicbook-vm:~/theepicbook$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
azureuser@epicbook-vm:~/theepicbook$ node server.js
going to html route
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Author'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM 'Author' FROM 'bookstore'
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Book'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM 'Book' FROM 'bookstore'
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Cart'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM 'Cart' FROM 'bookstore'
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Checkout'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM 'Checkout' FROM 'bookstore'
Executing (default): SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE' AND TABLE_NAME = 'Cartbook'
AND TABLE_SCHEMA = 'bookstore'
Executing (default): SHOW INDEX FROM 'Cartbook' FROM 'bookstore'
App listening on PORT 8080
Executing (default): SELECT 'Book'.id, 'Book'.title, 'Book'.genre, 'Book'.pubYear, 'Book'.price, 'Book'.inventory,
'Book'.bookDescription, 'Book'.createdAt, 'Book'.updatedAt, 'Book'.authorId, 'Author'.id AS 'Author.id', 'Author'.
firstName AS 'Author.firstName', 'Author'.lastName AS 'Author.lastName', 'Author'.createdAt AS 'Author.createdAt', 'Autho
r'.updatedAt AS 'Author.updatedAt' FROM 'Book' AS 'Book' LEFT OUTER JOIN 'Author' AS 'Author' ON 'Book'.authorId = 'Author
'.id LIMIT 9;
Executing (default): SELECT DISTINCT('genre') AS 'DISTINCT' FROM 'Book' AS 'Book';
Executing (default): SELECT count(*) AS 'count' FROM 'Cart' AS 'Cart';
Executing (default): SELECT 'Book'.id, 'Book'.title, 'Book'.genre, 'Book'.pubYear, 'Book'.price, 'Book'.inventory,
'Book'.bookDescription, 'Book'.createdAt, 'Book'.updatedAt, 'Book'.authorId, 'Author'.id AS 'Author.id', 'Author'.
firstName AS 'Author.firstName', 'Author'.lastName AS 'Author.lastName', 'Author'.createdAt AS 'Author.createdAt', 'Autho
r'.updatedAt AS 'Author.updatedAt' FROM 'Book' AS 'Book' LEFT OUTER JOIN 'Author' AS 'Author' ON 'Book'.authorId = 'Author
'.id LIMIT 9;
Executing (default): SELECT DISTINCT('genre') AS 'DISTINCT' FROM 'Book' AS 'Book';
Executing (default): SELECT count(*) AS 'count' FROM 'Cart' AS 'Cart';
Executing (default): SELECT 'id', title, genre, pubYear, price, inventory, bookDescription, createdAt, updatedAt,
authorId FROM 'Book' AS 'Book' WHERE 'Book'.id = '2';
Executing (default): INSERT INTO 'Cart' ('id', quantity, price, createdAt, updatedAt) VALUES (DEFAULT, ?, ?, ?, ?);
Executing (default): SELECT 'createdAt', 'updatedAt', 'BookId', 'CartId' FROM 'Cartbook' AS 'Cartbook' WHERE 'Cartbook'. 'CartI
d' = 1 AND 'Cartbook'. 'BookId' IN (2);
Executing (default): INSERT INTO 'Cartbook' ('createdAt', 'updatedAt', 'BookId', 'CartId') VALUES ('2025-10-04 10:52:58', '2025-10
-04 10:52:58', 2, 1);
Executing (default): SELECT 'Book'.id, 'Book'.title, 'Book'.genre, 'Book'.pubYear, 'Book'.price, 'Book'.inventory,
'Book'.bookDescription, 'Book'.createdAt, 'Book'.updatedAt, 'Book'.authorId, 'Author'.id AS 'Author.id', 'Author'.
firstName AS 'Author.firstName', 'Author'.lastName AS 'Author.lastName', 'Author'.createdAt AS 'Author.createdAt', 'Autho
r'.updatedAt AS 'Author.updatedAt' FROM 'Book' AS 'Book' LEFT OUTER JOIN 'Author' AS 'Author' ON 'Book'.authorId = 'Author
'.id LIMIT 9;
Executing (default): SELECT DISTINCT('genre') AS 'DISTINCT' FROM 'Book' AS 'Book';
Executing (default): SELECT count(*) AS 'count' FROM 'Cart' AS 'Cart';
Executing (default): SELECT 'id', title, genre, pubYear, price, inventory, bookDescription, createdAt, updatedAt
```

Screenshots of PM installation done during Troubleshooting:

```

azureuser@epicbook-vm:~/theepicbook$ sudo npm i -g pm2
added 133 packages in 14s
13 packages are looking for funding
  run `npm fund` for details
azureuser@epicbook-vm:~/theepicbook$ pm2 start server.js --name epicbook-api

```



Runtime Edition

PM2 is a Production Process Manager for Node.js applications with a built-in Load Balancer.

Start and Daemonize any application:
\$ pm2 start app.js

Load Balance 4 instances of api.js:
\$ pm2 start api.js -i 4

Monitor in production:
\$ pm2 monitor

Make pm2 auto-boot at server restart:
\$ pm2 startup

To go further checkout:
<http://pm2.io/>

```

[PM2] Spawning PM2 daemon with pm2_home=/home/azureuser/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /home/azureuser/theepicbook/server.js in fork_mode (1 instance)
[PM2] Done.

```

```

azureuser@epicbook-vm:~/theepicbook$ pm2 start server.js --name epicbook-api --watch
[PM2][ERROR] Script already launched, add -f option to force re-execution
azureuser@epicbook-vm:~/theepicbook$ pm2 start server.js --name epicbook-api --watch -f
[PM2] Starting /home/azureuser/theepicbook/server.js in fork_mode (1 instance)
[PM2] Done.

```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	epicbook-api	default	1.0.0	fork	12869	7s	12	online	0%	89.6mb	azu-	disabled
1	epicbook-api	default	1.0.0	fork	12890	0s	0	online	0%	13.6mb	azu-	enabled

```

azureuser@epicbook-vm:~/theepicbook$ ls
'Installation & Configuration Guide.md'  README.md  config  db  models  node_modules  package-lock.json  package.json  public  routes  server.js  views
azureuser@epicbook-vm:~/theepicbook$ pm2 status

```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	epicbook-api	default	1.0.0	fork	13114	7s	22	online	0%	67.1mb	azu-	disabled
1	epicbook-api	default	1.0.0	fork	13102	12s	9	online	0%	57.8mb	azu-	enabled

```

azureuser@epicbook-vm:~/theepicbook$ sudo systemctl restart nginx
azureuser@epicbook-vm:~/theepicbook$ curl -i http://127.0.0.1:3001/api/health
curl: (7) failed to connect to 127.0.0.1 port 3001 after 0 ms: Couldn't connect to server
azureuser@epicbook-vm:~/theepicbook$ curl -i http://127.0.0.1:3002/api/health
curl: (1) Received HTTP/0.9 when not allowed
azureuser@epicbook-vm:~/theepicbook$ ^C
azureuser@epicbook-vm:~/theepicbook$ pm2 status

```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	epicbook-api	default	1.0.0	fork	13927	8s	57	online	0%	62.0mb	azu-	disabled
1	epicbook-api	default	1.0.0	fork	13916	12s	44	online	0%	53.6mb	azu-	enabled

```

azureuser@epicbook-vm:~/theepicbook$ ^C
azureuser@epicbook-vm:~/theepicbook$ ls
'Installation & Configuration Guide.md'  README.md  config  db  models  node_modules  package-lock.json  package.json  public  routes  server.js  views
azureuser@epicbook-vm:~/theepicbook$ npm install
( [ ] ) i reify:fsevents: [111] reify mark deleted [ '/home/azureuser/theepicbook/node_modules/fsevents' ]
( [ ] ) i reify:fsevents: [111] reify mark deleted [ '/home/azureuser/theepicbook/node_modules/fsevents' ]
( [ ] ) i reify:fsevents: [111] reify mark deleted [ '/home/azureuser/theepicbook/node_modules/fsevents' ]
^C
( [ ] ) i reify:fsevents: [111] reify mark deleted [ '/home/azureuser/theepicbook/node_modules/fsevents' ]

```

LinkedIn Post1:

https://www.linkedin.com/posts/adrina-colaco_deploy-a-react-application-on-an-azure-virtual-activity-7379232209184104448-2jzT?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post2:

https://www.linkedin.com/posts/adrina-colaco_azure-devops-threetierapp-activity-7379239940330541056-PP_2?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post3:

https://www.linkedin.com/posts/adrina-colaco_azure-cloudmigration-staticwebsite-activity-7379262125208375296-BueQ?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post4:

https://www.linkedin.com/posts/adrina-colaco_overcoming-challenges-through-collaboration-activity-7380235245821886464-FQOA?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

Blogs:

<https://medium.com/@adcola13/deploy-a-react-application-on-an-azure-virtual-machine-ba79f4d8d071>

The End

P.S. This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by [Pravin Mishra](#). You can start your DevOps journey for free from his [YouTube Playlist](#).